THE SQUARE-ROOT-OF-TIME RULE ON ONE PAGE

Spread grows like $\sqrt{n}$, told with dice

Add up many independent rolls and the total drifts in proportion to n while its scatter grows only as $\sqrt{n}$.

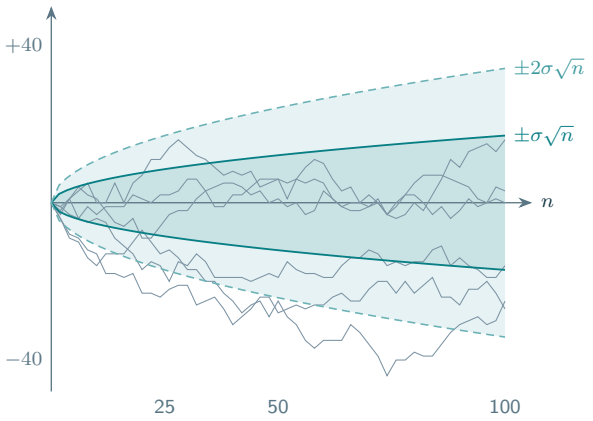
THE RULE

Variances add, so standard deviations go as $\sqrt{n}$

One fair die has mean 3.5 and variance $\frac{35}{12}$, so its standard deviation is $\sigma = \sqrt{35/12} \approx 1.71$. For independent rolls the variances add, and the total of n rolls has

$$\text{mean} \;=\; 3.5\,n, \qquad \text{SD} \;=\; \sigma\sqrt{n} \;\approx\; 1.71\sqrt{n}$$

Quadrupling the number of rolls doubles the spread. Subtracting 3.5 from each roll leaves a fair random walk whose typical distance from zero after n steps is $1.71\sqrt{n}$; the fan beside it shows six such walks of 100 rolls inside bands at one and two standard deviations.



Running total of (roll − 3.5) for six walks; the bands widen as $\sqrt{n}$, never as n.

LONG-RUN DICE

The total spreads, the average settles

rolls n	mean total	SD of total	95% of totals in	SD of average	spread ÷ mean
1	3.5	1.71	1 to 6, all	1.71	49%
4	14	3.42	7.3 to 20.7	0.85	24%
16	56	6.83	42.6 to 69.4	0.43	12%
100	350	17.1	316.5 to 383.5	0.17	4.9%
400	1400	34.2	1333 to 1467	0.085	2.4%

Every factor of 4 in rolls doubles the spread of the total and halves the spread of the average, which is the law of large numbers seen at its proper rate. By the central limit theorem the totals pile into a bell curve whatever the shape of one die, which is what lets ± 1.96 SD capture 95%.

FROM DICE TO PRICES

Daily log returns play the part of rolls

If each day's log return is an independent draw with standard deviation σ_{day}, the log return over T days is a sum of T draws, so

$$\sigma_T = \sigma_{\text{day}}\sqrt{T}, \qquad \sigma_{\text{year}} \approx \sigma_{\text{day}}\sqrt{252},$$

using 252 trading days. A typical daily move of 1% becomes about 2.2% a week, 4.6% a month and 15.9% a year. Any steady drift μ grows in proportion to T, so it overtakes one standard deviation of scatter only after $T = (\sigma/\mu)^2$, about five years for $\mu = 7\%$ and $\sigma = 15.9\%$ a year.

horizon	days	σ
day	1	1.0%
week	5	2.2%
month	21	4.6%
year	252	15.9%

Scaling a 1% daily standard deviation by $\sqrt{T}$.

WHY VARIANCES ADD

Independent steps are at right angles

Squaring a sum of deviations gives the squares plus cross terms $2\,d_i d_j$, and for independent rolls each cross term averages to zero, leaving $\operatorname{Var}\left(\sum d_i\right) = \sum \operatorname{Var}(d_i)$. Independent steps combine like perpendicular sides in Pythagoras, so n steps of size σ reach $\sigma\sqrt{n}$, the diagonal of an n-dimensional cube; the standard deviation is the RMS of the deviation, the same mean that measures alternating current.

WHEN THE RULE BREAKS

Real prices are not quite dice

The rule needs independent, identically distributed steps. Momentum makes long-horizon spread grow faster than $\sqrt{T}$ and mean reversion slower; volatility comes in clusters, so a calm month predicts a calm month; and daily returns have fatter tails than a bell curve, with crashes far more frequent than $\pm 1.96\,\sigma$ suggests. The $\sqrt{T}$ rule is the benchmark against which these departures are measured, and it is not investment advice.

Conventions. SD is standard deviation, variance is its square, T counts trading days, and log returns are used so that returns over successive days add.

Constructed by Silicon · Anthropic Claude — Architect Carbon · LeeMcCullochJames